

Automatic Detection of Illicit Activity in Bitcoin: From Ego-Network Transfer Learning to Full-Graph Temporal Models

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Abstract—Social interactions exhibit persistent structural regularities, making them a natural substrate for learning and generalization in graph-based systems. In cryptocurrency transaction networks, these regularities manifest as recurring local interaction patterns associated with illicit activity.

In this work, conducted within the TransGraph project, we study the automatic detection of illicit behavior in the Bitcoin transaction graph from two complementary perspectives: (i) localized structural learning via ego-networks, and (ii) scalable full-graph analysis using diffusion and temporal methods.

At the local level, we show that illicit activity is embedded in cohesive structures such as chains, star configurations, and triangular motifs. To capture these patterns, we employ expressive graph neural networks with 3-Weisfeiler–Lehman (3-WL) power and demonstrate that higher-order expressiveness significantly improves recall in a strict pairwise transfer setting across ego-networks.

At the global level, we evaluate scalable approaches, including Personalized PageRank, temporal diffusion methods, and Temporal Graph Networks (TGN), a state-of-the-art model for dynamic graphs. Despite their scalability, these methods fail to achieve a satisfactory balance between precision and recall: diffusion-based methods produce many false positives, while TGN achieves high precision but misses most illicit nodes.

These results reveal a fundamental limitation of full-graph temporal and diffusion-based approaches and suggest that illicit activity detection is primarily driven by stable local structural patterns rather than global temporal dynamics.

Overall, our study highlights a key trade-off between expressiveness and scalability, and points toward structure-aware, higher-order models as a promising direction for scalable fraud detection in large financial networks.

Index Terms—Transfer learning, graph neural networks, provably powerful graph networks, Bitcoin fraud detection, cryptocurrency transaction graphs, large-scale graph learning, full-graph analysis, scalability, diffusion-based methods, PageRank, temporal graph networks, semi-supervised learning

I. INTRODUCTION

“*Man is by nature a social animal.*”

— Aristotle [1]

Social interactions play a fundamental role in shaping complex systems. Across different domains, individuals communicate, coordinate, and establish trust through recurring patterns of interaction, giving rise to stable local social structures. Because these structures often persist even when the surrounding

environment changes, they provide a natural candidate for transferable knowledge.

This observation motivates the central question of this work: **can structurally similar interaction patterns be exploited for transfer learning across graphs?**

Graph-structured data provide a natural representation of such interactions, yet learning models that generalize across graphs with differing topologies remains limited [2], [3]. This challenge is particularly important in applications where graphs evolve, differ structurally, or are observed only partially [4]. Cryptocurrency transaction networks provide a realistic and especially difficult testbed: they are large, weakly labeled, structurally heterogeneous, and inherently temporal [5].

This work is conducted as part of the TransGraph project, led by Konstantin Avrachenkov (Inria, NEO team) in collaboration with Arnaud Legout (Inria, DIANA team). The project aims to develop scalable and transferable graph learning methods by combining invariant and equivariant GNN architectures with motif- and subgraph-centric representations. Within this framework, we study fraud detection in Bitcoin transaction graphs.

Our starting point is the hypothesis that illicit activity is not merely a node-level phenomenon, but is often embedded in recurring *local structural patterns*. We therefore focus first on *ego-networks*, where localized subgraphs serve as portable units of information. In this setting, motifs and small subgraphs are treated as fundamental units of representation. Recurring transactional configurations such as chains, hub-and-spoke structures, and triangular motifs are extracted from labeled subgraphs and used to study whether structural information can transfer across graph instances.

To evaluate this hypothesis, we consider a strict *pairwise transfer* setting, where a model is trained on one ego-network and tested on another. This allows us to analyze how well structural patterns generalize across distinct local graphs. We further examine the role of model expressiveness in this transfer scenario by comparing models of increasing structural power, from standard message-passing architectures to more expressive higher-order GNNs.

This leads to the first research direction of the paper:

motif-centric transfer learning for Bitcoin fraud detection. The second direction concerns the role of model complexity in transferability, namely whether more expressive nonlinear architectures generalize better than lighter propagation-based models across ego-networks.

However, ego-network analysis alone is not sufficient for realistic deployment. The Bitcoin transaction graph contains tens of millions of nodes and edges, making highly expressive GNNs computationally infeasible at the full-graph level. This motivates the second part of our study, where we investigate scalable methods that operate directly on the entire graph, including diffusion-based and temporal approaches.

In this setting, we evaluate methods such as Personalized PageRank, Temporal PageRank, and Temporal Graph Networks (TGN), as well as temporal embedding approaches, to assess their ability to capture illicit activity at scale.

Formally, we consider a binary node classification task, where nodes are classified as *licit* or *illicit*. The task is studied under two complementary settings: (i) transfer learning across ego-networks, focusing on localized structural generalization, and (ii) scalable full-graph learning, focusing on diffusion- and temporal-based methods.

Overall, this work provides a unified analysis of structural and temporal approaches to illicit activity detection in large-scale transaction graphs.

The code and implementation details are publicly available at: <https://github.com/ZeevWeizmann/Automatic-Detection-of-Illicit-Activity-in-Bitcoin>.

A. Contributions

This work makes the following contributions:

- **Structural perspective on transfer in transaction graphs.** We investigate recurring local interaction patterns in cryptocurrency transaction networks as transferable units for learning, highlighting socially meaningful structures such as chains, hubs, and triadic motifs. Our results indicate that capturing such patterns requires higher-order expressiveness beyond standard message-passing models.
- **Comparative analysis of graph expressiveness.** We systematically study models of increasing expressive power, corresponding to different levels of the Weisfeiler–Lehman hierarchy (1-WL, 2-WL, and 3-WL), and evaluate their behavior in a transfer learning setting. We show that higher-order models improve the detection of structurally coordinated patterns, particularly in terms of recall.
- **Ego-network transfer learning on blockchain data.** We construct ego-networks from a large-scale Bitcoin transaction graph using a BFS-based procedure and introduce a pairwise transfer setting to evaluate structural generalization across heterogeneous subgraphs.
- **Bridging local and global graph learning.** We investigate how structural insights obtained from ego-network modeling relate to scalable learning on the full transaction

graph, highlighting the challenges of transferring localized structural patterns to large-scale settings.

- **Scalable evaluation on full-graph methods.** We conduct an empirical evaluation of diffusion-based and temporal approaches on the full Bitcoin transaction graph, including Personalized PageRank, Temporal PageRank, Temporal Graph Networks, and temporal embedding methods.
- **Insights on structural vs. temporal signals.** Our results suggest that localized structural patterns play a central role in illicit activity detection, while temporal and global diffusion signals alone may not fully capture the underlying organization of such behavior.

II. THE DATASET

The study relies on a large-scale Bitcoin transaction graph constructed from raw blockchain data. The resulting directed graph contains approximately 31,535,968 nodes and 34,769,058 edges, corresponding to individual transactions, and includes 235,030 labeled addresses annotated as either *licit* or *illicit*.

To enable scalable analysis while focusing on local interaction patterns, we extract ego-networks from the global transaction graph. This process relies on a *modified breadth-first search (BFS)* procedure, specifically adapted to operate efficiently on large-scale blockchain data and to support ego-network extraction.

Unlike a standard BFS, the procedure was extended to preserve transaction directionality, maintain contextual information about interactions, and ensure consistent extraction of localized subgraphs around target nodes. This adaptation allows the method to operate as a core building block for constructing ego-networks in a scalable manner.

Using this procedure, k -hop induced subgraphs are constructed around labeled addresses, with $k = 1$, yielding ego-network representations centered on individual entities. These ego-networks capture the immediate transactional neighborhood of each address while avoiding the computational cost of full-graph processing.

In addition to structural information, we incorporate a notion of temporal ordering by retrieving the block height associated with each transaction. This provides a proxy for time, allowing transactions to be positioned along the blockchain timeline. However, the BFS traversal itself does not enforce chronological constraints, and therefore the constructed graph does not strictly follow temporal order. As a result, the dataset can be viewed as a *pseudo-temporal* graph, where temporal information is available as node and edge attributes but is not used to constrain graph construction.

Despite this approximation, temporal models are evaluated to assess whether temporal signals alone can compensate for the absence of explicit structural modeling. We adopt this approximation as a practical compromise between preserving structural connectivity and incorporating temporal context.

Address labels are provided by researchers at the University Mohammed VI Polytechnic (UM6P), who specialize in cryptocurrency intelligence and illicit activity detection. The

labeling process combines multiple heterogeneous sources, including publicly available databases of fraudulent addresses and information extracted from darknet and underground forums. To process this unstructured data, large language models (LLMs) are used to extract and normalize cryptocurrency addresses mentioned in textual content, following recent advances in underground forum analysis [6]. These labeled addresses serve as anchors for ego-network extraction and subsequent graph-based learning.

For comparison, widely used benchmark datasets provide substantially more limited supervision. For example, the dataset introduced in [7] contains only a few thousand labeled illicit addresses and a large proportion of unlabeled nodes. In contrast, the scale and density of labeled data in our setting enable a more detailed investigation of local structural patterns and their role in transferable graph learning.

III. EXPRESSIVE GNNs FOR EGO-COMMUNITY TRANSFER

The empirical analysis of ego-networks extracted from the Bitcoin transaction graph revealed recurring and highly structured local patterns associated with illicit activity. In particular, we repeatedly observed small, socially cohesive configurations such as chains, star-like structures, and densely connected ego-networks containing triangular motifs. These patterns emerged naturally during graph exploration rather than being imposed by modeling assumptions.

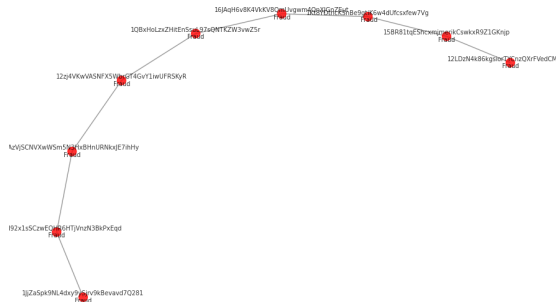


Fig. 1. Chain-like ego-network composed exclusively of fraud-labeled addresses. Such structures correspond to sequential fund transfers and are commonly used to increase transaction depth and reduce traceability in illicit activity.

Figure 1 illustrates a typical chain structure formed by fraud-labeled addresses. This pattern reflects multi-hop transaction strategies, where funds are passed through a sequence of addresses to obscure their origin. While simple in appearance, such chains represent coordinated behavior that is difficult to capture using purely local node features.

Another recurrent configuration is the star-like ego-network shown in Figure 2. Here, a single central node interacts with multiple neighbors, suggesting a service-like role such as reward distribution, coordination, or laundering. These star-shaped patterns further support the hypothesis that illicit activity is organized around localized social roles rather than isolated addresses.

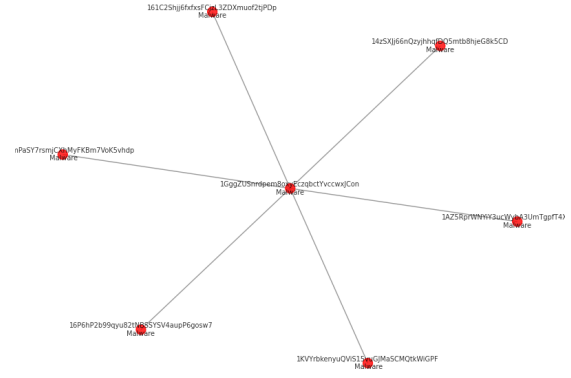


Fig. 2. Star-shaped ego-network centered around a malware-labeled address. This topology is indicative of centralized services distributing funds to multiple peripheral nodes, a pattern frequently observed in illicit ecosystems.

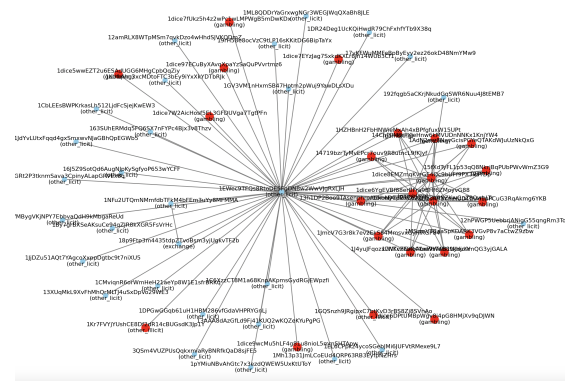


Fig. 3. Dense ego-network centered on a gambling-related address. The presence of multiple triangular motifs and tightly connected neighbors suggests coordinated activity and highlights the importance of higher-order structural patterns in transaction graphs.

Most notably, we observed dense ego-networks containing multiple triangles, as illustrated in Figure 3. Such triangular motifs indicate triadic closure and strong local cohesion, suggesting coordinated interaction among participants. These observations naturally raise the question of whether standard graph neural networks are sufficiently expressive to detect and exploit such structures.

Beyond our empirical observations, triangular structures have also been identified as fundamental motifs in prior studies of financial transaction networks. In particular, Liu et al. [8] systematically analyzed temporal motifs in large-scale payment platforms such as Mercari, Venmo, and internal J.P. Morgan Chase transaction data, and showed that triadic interaction patterns occur frequently [9] and carry strong semantic meaning related to coordinated financial behavior. Their results indicate that triangles are not only static structural motifs, but also persist as meaningful temporal patterns in real-world financial networks. This independent evidence further supports the relevance of triangular motifs as a transferable and socially meaningful unit of analysis in fraud detection

and financial graph learning.

Our perspective was further informed by classical sociological theories. Georg Simmel [10] and Mark Granovetter [11] emphasized the fundamental role of triadic relationships in the formation of trust and social stability. Their core insight is that if actor A trusts B and B trusts C , then A is more likely to trust C , making triangles a basic unit of social cohesion. This concept of embeddedness suggests that detecting triangular motifs is crucial for identifying tightly coordinated communities.

This view is further supported by empirical studies of covert social structures. For instance, Tsvetov and Kouznetsov [12] report that terrorist cells often exhibit densely connected triadic structures, where members are linked through strong ties of trust, loyalty, and shared identity. Such configurations, sometimes forming overlapping closed triads or small cliques, reinforce internal cohesion while limiting exposure to external actors. These findings highlight that triangular motifs are not only abstract structural patterns, but also fundamental organizational units in real-world clandestine networks.

This convergence across domains suggests that illicit financial behavior, including money laundering in cryptocurrency networks, may follow similar organizational principles as other covert systems. In this perspective, triangular motifs can be interpreted as universal building blocks of coordinated and concealed activity.

This idea also resonates with Harrison White’s framework of identity and control [13], where structurally equivalent positions and tightly knit clusters play a central role in sustaining social coherence. Although not formulated in graph-theoretic terms, White’s work implicitly highlights the importance of triadic configurations in the emergence of social meaning and collective behavior.

These sociological principles provide a theoretical foundation for treating triadic motifs as transferable units of social organization in graph-based learning.

These theoretical insights align naturally with highly expressive graph neural network architectures. Standard message-passing GNNs, limited to the expressive power of the 1-WL test [2], cannot reliably distinguish many of the structures observed in [3] Figures 1–3. In contrast, 3-WL-level models such as Provably Powerful Graph Networks (PPGNs) are theoretically capable of capturing triangles, cliques, and higher-order motifs, making them particularly well suited for modeling socially cohesive ego-communities.

Building on the concept of *egomunities* introduced by Friggeri et al. [14], defined as person-based, locally dense and triad-rich communities centered around an individual (ego), we treat these socially cohesive structures as transferable entities across ego-networks. By explicitly modeling and aligning their internal motif structure, expressive GNNs can leverage egomunities to enable robust transfer learning in socially rich fraud detection settings.

IV. SCALABLE AND TEMPORAL METHODS FOR FULL-GRAPH ANALYSIS

A central challenge in the proposed direction is the computational cost associated with highly expressive graph neural networks. Naive approaches to achieving 3-WL-level expressiveness rely on explicit third-order tensor representations, leading to prohibitive memory and time complexity proportional to $O(n^3)$. This creates a fundamental trade-off between expressiveness and scalability, particularly when working with large collections of ego-networks.

While highly expressive models are theoretically well suited for capturing triangular motifs and higher-order social interactions, their practical applicability becomes severely constrained for large-scale graphs. This limitation is especially pronounced in the context of the Bitcoin transaction network, which contains tens of millions of nodes and edges. Even with access to high-memory computing resources (up to 1 TB of RAM), scaling expressive models to the full graph remains infeasible.

To address this limitation, we investigate alternative approaches that are designed to operate directly on the full graph under realistic computational constraints. Unlike expressive GNNs, these methods rely on diffusion, propagation, or optimization principles rather than deep neural architectures.

In particular, we consider graph-based semi-supervised learning methods within the generalized optimization framework proposed by Avrachenkov et al. [15]. These methods can be interpreted as infinitely deep linear message-passing schemes without learnable parameters, enabling efficient inference on graphs with millions of nodes.

We further investigate diffusion-based methods such as Personalized PageRank, which propagate label information through the graph structure. To incorporate temporal information, we consider extensions such as Temporal PageRank, where propagation is modulated by time-dependent decay, allowing recent interactions to have greater influence.

In addition, we explore temporal graph representations where the Bitcoin transaction network is modeled as a sequence of time-stamped events. In this setting, we consider approaches combining temporal diffusion with learned representations [16], including Temporal Graph Neural Networks (TGN) [5] and the Zebra framework, which uses temporal Personalized PageRank to construct time-aware node embeddings from the most influential temporal neighbors. In our setting, these embeddings are used as input features for a downstream classifier [16].

These approaches are not intended as simple baselines, but rather as a complementary paradigm to expressive GNNs. While expressive models focus on capturing higher-order structural patterns in local ego-networks, scalable and temporal methods enable analysis at the full-graph level.

Our goal is to evaluate the extent to which these approaches can capture illicit activity patterns in large-scale transaction graphs, and to compare their effectiveness with structurally expressive models. In particular, we aim to assess whether

temporal signals and global diffusion processes provide additional predictive power beyond local structural motifs.

V. PROVABLY POWERFUL GRAPH NETWORKS

As the main high-expressiveness modeling framework of this work, we rely on Provably Powerful Graph Networks (PPGNs) [17], [18]. PPGNs form a class of invariant and equivariant graph neural networks whose expressive power matches the 3-dimensional Weisfeiler–Lehman (3-WL) test, thereby overcoming the fundamental limitations of standard message-passing GNNs, which are bounded by the 1-WL test [19].

Importantly, PPGNs do not explicitly operate on full third-order tensors. Instead, they achieve 3-WL-level expressiveness while primarily manipulating second-order representations of size $n \times n$. By enforcing permutation invariance at the architectural level, PPGNs avoid redundant representations of equivalent node configurations, allowing them to capture higher-order interactions at a significantly lower computational cost than explicit tensor-based methods.

The internal computation of PPGNs is based on structured block operations combining multiple learnable transformations and bilinear interactions, as illustrated in Figure 4. This block structure enables the model to simulate higher-order interactions without explicitly constructing higher-order tensors.

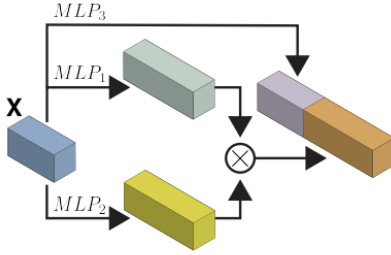


Figure 2: Block structure.

Fig. 4. Block structure of a Provably Powerful Graph Network (PPGN). The architecture operates on second-order representations and combines multiple MLP transformations with bilinear interactions, enabling the model to capture higher-order structural patterns such as triangles and cliques while preserving permutation invariance.

The expressive power of PPGNs increases with the number of stacked blocks. A single block captures pairwise interactions between nodes, corresponding approximately to 2-WL-level expressiveness. Stacking multiple blocks allows the composition of interactions: in particular, using two blocks enables the model to represent third-order dependencies and reach 3-WL-level expressiveness.

This distinction is crucial in the context of fraud detection. Many illicit behaviors are not encoded at the level of individual edges, but rather emerge as higher-order patterns such as chains, hubs, and especially triangular structures. Triangles represent a minimal unit of coordinated interaction involving three entities and are therefore inherently invisible to models limited to pairwise aggregation.

We adopt PPGNs as the core expressive backbone for ego-network analysis precisely because they enable the detection of such higher-order patterns. Empirically, we observe that models with 3-WL-level expressiveness significantly improve the detection of illicit nodes compared to 1-WL baselines.

This improvement can be explained by the ability of PPGNs to capture structural motifs underlying coordinated behavior. While standard GNNs often achieve competitive overall accuracy, they tend to miss illicit nodes that are embedded in complex local structures. In contrast, PPGNs are able to identify these patterns, leading to a substantial increase in recall for the illicit class.

To further support transfer learning across ego-networks, we augment the PPGN backbone with an embedding alignment mechanism. Following recent work on graph transfer learning [4], [20], we employ a Sinkhorn-based optimal transport formulation [20] to align latent representations across graphs with differing topologies [4]. This alignment operates at the level of learned embeddings and enforces consistency between structurally similar ego-communities.

The effectiveness of the proposed approach is evaluated under a pairwise transfer learning protocol, where models are trained on one ego-network and evaluated on another. This setting provides a strict test of generalization across structurally distinct graphs.

In the context of fraud detection, the primary challenge lies in identifying illicit nodes that exhibit subtle or weakly observable patterns [7]. We therefore focus on recall for the illicit class as a key evaluation metric, as it reflects the ability of the model to uncover coordinated and hard-to-detect fraudulent behavior that remains missed by less expressive architectures.

VI. PAIRWISE TRANSFER LEARNING IN EGO-NETWORKS

To evaluate the transferability of structural representations across graphs, we consider a pairwise transfer learning setting based on ego-networks extracted from the Bitcoin transaction graph.

Each ego-network is constructed as a k -hop subgraph (with $k = 2$) centered around a labeled address. This choice enables the capture of both immediate neighbors and second-order interactions, which are essential for modeling structural patterns such as chains, star configurations, and triangular motifs.

These ego-networks are treated as independent graph instances, allowing us to simulate distribution shifts between training and test data.

The transfer setting is illustrated in Figure 5, where a model trained on a source ego-network is applied to a structurally different target ego-network.

In this setting, a model is trained on one ego-network and evaluated on another, resulting in a strict transfer learning scenario where both graph topology and node identities differ between training and test instances.

We evaluate models of increasing expressive power corresponding to different levels of the Weisfeiler–Lehman hierar-

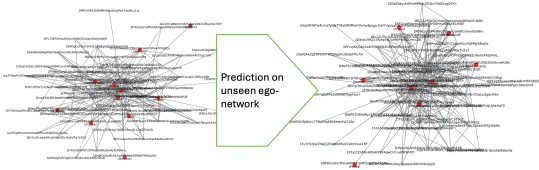


Fig. 5. Pairwise transfer learning between ego-networks. A model is trained on a source ego-network and applied to a structurally distinct target ego-network, testing its ability to transfer learned structural patterns across graphs.

chy. The results, averaged across multiple train/test pairs, are summarized in Table I.

TABLE I
AVERAGE PERFORMANCE ACROSS PAIRWISE TRANSFER EXPERIMENTS

Model	Precision	Recall	F1-score
PPGN (3-WL)	0.15	0.83	0.26
PairGNN (2-WL)	0.20	0.52	0.25
GCN (1-WL) [21]	0.62	0.78	0.64

Importantly, while GCN achieves the best overall balance, it relies primarily on local aggregation and may fail to capture coordinated patterns in more complex settings, whereas PPGN explicitly targets higher-order structures and achieves significantly higher recall.

The results reveal a fundamental trade-off between expressiveness and transferability. The GCN baseline (1-WL) [21] provides the most balanced performance, achieving relatively high precision and recall, suggesting that simple local aggregation captures robust signals that generalize well across ego-networks.

The PairGNN (2-WL) model does not consistently improve performance, indicating that pairwise interactions alone are insufficient to capture transferable structural information.

In contrast, the PPGN (3-WL) model achieves very high recall, often detecting most illicit nodes. This confirms that higher-order expressiveness enables the model to capture complex structural patterns associated with coordinated behavior.

However, this gain in recall comes at the cost of very low precision, resulting in a large number of false positives. This behavior suggests that highly expressive models tend to overfit structural motifs and generalize them too aggressively across graphs. As a result, they detect many structures that resemble illicit patterns, but fail to distinguish between true and spurious occurrences.

Another important observation is the high variability across different train/test pairs. In several cases, all models fail completely, highlighting the inherent difficulty of transfer learning across ego-networks with heterogeneous structural properties.

Overall, these results indicate that higher-order expressiveness improves recall but does not guarantee robustness, and may lead to over-detection of illicit activity when structural patterns are not fully transferable across graphs.

While these findings highlight important limitations of pairwise transfer learning, they also point to a potential direction

for improvement. In particular, the ego-networks considered in this study are relatively small, which may limit the ability of highly expressive models to fully exploit higher-order structural patterns.

We hypothesize that larger ego-networks, containing richer motif structures and more complex interaction patterns, would provide a more suitable setting for expressive architectures such as PPGNs. However, scaling the current approach to larger ego-networks remains computationally challenging due to the quadratic complexity of second-order representations.

Exploring transfer learning on larger ego-networks therefore represents an important direction for future work, where higher-order models are expected to demonstrate stronger advantages over less expressive baselines.

VII. FULL-GRAPH LEARNING: DIFFUSION AND TEMPORAL APPROACHES

While expressive GNNs demonstrate strong performance on ego-networks, their application to the full Bitcoin transaction graph is computationally infeasible. Indeed, even memory-efficient implementations of higher-order models such as PPGNs require prohibitive resources when scaling to graphs with tens of millions of nodes. As illustrated in our experiments, applying such models to the full graph would require several thousand terabytes of memory, making this approach impractical.

This limitation motivates a shift toward scalable methods that operate directly on the full graph. In particular, we investigate diffusion-based approaches, temporal extensions of PageRank, and temporal graph learning methods.

A. Diffusion-Based Semi-Supervised Learning

We begin with the generalized optimization framework for graph-based semi-supervised learning proposed by Avrachenkov et al. [15]. This approach formulates label propagation as an iterative diffusion process over the graph:

$$F_{t+1} = (1 - \alpha)Y + \alpha D^{-\sigma} W D^{\sigma-1} F_t \quad (1)$$

where Y represents initial labels (seed information), W is the adjacency matrix, and D is the degree matrix.

This framework can be interpreted as an infinite-depth linear message passing scheme, allowing efficient propagation of label information over very large graphs without requiring explicit model training.

Importantly, this approach scales to the full Bitcoin graph and does not require storing high-order tensor representations, making it suitable for large-scale deployment.

B. Personalized PageRank and Seed Selection

Building on diffusion-based methods, we evaluate Personalized PageRank (PPR) [22] under different seed initialization strategies.

The results are summarized in Table II.

The results highlight a strong dependence of PPR performance on the choice of seed nodes.

TABLE II
IMPACT OF SEED SELECTION ON PPR PERFORMANCE

Method	Precision	Recall	F1-score	Accuracy
Degree seeds	0.143	0.530	0.225	0.780
Global PageRank	0.084	0.030	0.044	0.922
Personalized PR	0.054	0.023	0.033	0.917
Random seeds	0.133	0.513	0.211	0.769

Degree-based and random seed selection strategies achieve substantially higher recall (above 50%) compared to PageRank-based approaches. This suggests that illicit activity tends to be organized in localized regions of the graph, where diffusion from nearby nodes is effective in propagating signals.

Interestingly, qualitative inspection of the graph reveals that many of these localized regions are situated around known high-activity services, such as Bitcoin gambling platforms (e.g., SatoshiDice). These services act as hubs generating dense transactional neighborhoods, within which illicit addresses may be embedded or closely connected. As a result, diffusion processes initialized in such regions are more likely to reach illicit nodes, explaining the improved recall observed for degree-based and random seeding strategies.

In contrast, global PageRank and standard Personalized PageRank yield very low recall, despite achieving high accuracy. This discrepancy is explained by the extreme class imbalance: these methods tend to favor the majority (licit) class and fail to propagate sufficient signal toward rare illicit nodes.

Overall, diffusion-based approaches demonstrate a clear limitation: while they can propagate information efficiently over large graphs, they lack structural selectivity. As a result, they either diffuse signals too broadly (leading to low precision) or fail to reach relevant regions (leading to low recall).

These findings indicate that purely diffusion-based methods are insufficient to capture the complex structural patterns underlying illicit activity, reinforcing the need for models that explicitly encode higher-order interactions.

C. Temporal PageRank

To incorporate temporal dynamics, we extend Personalized PageRank (PPR) to Temporal PageRank (t-PPR) [23], [24]. This formulation integrates time into the random-walk process by introducing time-aware transition dynamics over a stream of interactions.

Unlike standard PageRank, t-PPR operates on a sequence of time-stamped edges, updating node importance scores incrementally as new interactions arrive. This streaming formulation allows the method to scale to large graphs without requiring storage of the full temporal structure.

In this work, we adopt a streaming variant inspired by Rozenshtein and Gionis [23], where node scores are updated online as transactions are processed.

The algorithm is presented in Algorithm 1.

Here, $r(u)$ represents the temporal PageRank score of node u , while $s(u)$ stores the residual probability mass propagated

Algorithm 1 Temporal PageRank (stream processing)

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1: Input: stream of edges  $E$ , transition probability  $\beta \in (0, 1]$ , teleportation probability  $\alpha$ 
2: Initialize  $r(u) = 0, s(u) = 0$  for all nodes  $u$ 
3: for each interaction  $(u, v, t) \in E$  do
4:    $r(u) \leftarrow r(u) + (1 - \alpha)$ 
5:    $s(u) \leftarrow s(u) + (1 - \alpha)$ 
6:    $r(v) \leftarrow r(v) + s(u)\alpha$ 
7:   if  $\beta < 1$  then
8:      $s(v) \leftarrow s(v) + s(u)(1 - \beta)\alpha$ 
9:      $s(u) \leftarrow s(u)\beta$ 
10:  else
11:     $s(v) \leftarrow s(v) + s(u)\alpha$ 
12:     $s(u) \leftarrow 0$ 
13:  end if
14: end for
15: Normalize  $r$ 
16: return  $r$ 

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through the interaction stream. The algorithm processes the transaction graph as a sequence of time-stamped edges and incrementally updates node scores.

Temporal PageRank introduces a decay parameter β controlling how much probability mass propagates over time:

- $\beta < 1$: emphasizes recent interactions
- $\beta = 1$: reduces to standard PageRank

This mechanism allows the model to prioritize recent transactions, which is particularly relevant in financial systems where behavior evolves over time.

a) *Temporal PageRank as diffusion.*: In the first setting, Temporal PageRank is used as a diffusion process, similar to Personalized PageRank, where a set of seed nodes is used to propagate information across the graph.

In this configuration, the method achieves moderate recall (approximately 0.5), indicating that it is able to detect a substantial fraction of illicit nodes. However, precision remains low (around 0.15), resulting in a large number of false positives. The overall accuracy is approximately 0.80.

This behavior suggests that temporal diffusion is effective at spreading signals within active regions of the graph, but lacks structural selectivity. In particular, diffusion tends to propagate signals broadly across dense and high-activity regions, including areas surrounding major transaction hubs such as Bitcoin gambling services (e.g., SatoshiDice), where many nodes are structurally close but not necessarily illicit.

b) *Personalized Temporal PageRank.*: In the second setting, we extend Temporal PageRank to a fully personalized formulation, where class labels are encoded as one-hot vectors and injected directly into both the accumulated rank (r) and the propagating mass (s). This allows the model to incorporate class information during diffusion and to operate as a classifier.

However, this approach performs significantly worse. The recall drops to approximately 0.06, precision to 0.05, and the F1-score to around 0.06, despite a relatively high accuracy (around 0.88), which is primarily driven by class imbalance.

TABLE III
PERFORMANCE OF PERSONALIZED TEMPORAL PAGERANK

Metric	Precision	Recall	F1-score	Accuracy
Illicit class	0.054	0.063	0.058	0.88

TABLE IV
CONFUSION MATRIX FOR PERSONALIZED TEMPORAL PAGERANK

	Predicted licit	Predicted illicit
True licit	140467	10650
True illicit	9058	611

This result indicates that introducing temporal dynamics in a fully personalized setting may dilute useful structural signals. Instead of improving discrimination, the model tends to over-smooth class information across the graph, leading to poor detection of illicit nodes.

Overall, these results suggest that temporal information alone is insufficient to capture the structural patterns underlying illicit activity.

While temporal diffusion can propagate signals effectively, it does not encode higher-order structural interactions such as triangles and dense ego-communities. In fact, introducing temporal dynamics may even degrade performance when it over-smooths informative structural signals.

This observation further supports the central hypothesis of this work: illicit activity is primarily a structural phenomenon rather than a purely temporal one.

D. Temporal Graph Learning: Zebra Framework

To further incorporate temporal dynamics and representation learning, we explore temporal graph neural networks combined with temporal diffusion using the Zebra framework.

Zebra operates on a stream of transactions and integrates three components:

- Temporal Personalized PageRank (T-PPR)
- Temporal node embeddings (Zebra encoder)
- A downstream classifier

This approach enables modeling both structural and temporal dependencies in the transaction graph by combining diffusion-based propagation with learned representations.

The core of the temporal diffusion component is based on a streaming top- k Temporal PageRank algorithm, known as SANTA, which maintains an approximation of the most relevant nodes over time.

The algorithm is presented in Algorithm 2.

This streaming formulation allows Zebra to process large-scale transaction graphs efficiently while maintaining temporal information. However, it requires substantial computational resources due to its in-memory processing of large interaction streams.

We evaluate the Zebra framework on approximately 10 million interactions, resulting in around 8 million temporal embeddings.

Algorithm 2 Single-scan top- k T-PPR (SANTA)

```

1: Input:  $\{(\tilde{\pi}_{i,t-}, m_{i,t-}) \mid i \in V\}$ , new interaction  $(v_i, v_j, e_{ij}, t)$ , threshold  $k$ 
2: Output: Approximated top- $k$  T-PPR dictionaries
3:  $\tilde{\pi}_{i,t+} \leftarrow \frac{m_{i,t-} - \beta}{m_{i,t-} - \beta + \beta} \tilde{\pi}_{i,t-} + \frac{\beta(1-\alpha)}{m_{i,t-} - \beta + \beta} (\tilde{\pi}_{j,t-} + \alpha \mathbf{1}_{j,t})$ 
4: Keep only top- $k$  nodes in  $\tilde{\pi}_{i,t+}$ 
5: if  $v_i \neq v_j$  then
6:    $\tilde{\pi}_{j,t+} \leftarrow \frac{m_{j,t-} - \beta}{m_{j,t-} - \beta + \beta} \tilde{\pi}_{j,t-} + \frac{\beta(1-\alpha)}{m_{j,t-} - \beta + \beta} (\tilde{\pi}_{i,t-} + \alpha \mathbf{1}_{i,t})$ 
7:   Keep only top- $k$  nodes in  $\tilde{\pi}_{j,t+}$ 
8: end if
9:  $m_{j,t+} \leftarrow m_{j,t-} - \beta + \beta$ 
10:  $m_{i,t+} \leftarrow m_{i,t-} - \beta + \beta$ 
11: return  $\tilde{\pi}_{i,t+}, \tilde{\pi}_{j,t+}$ 

```

TABLE V
ZEBRA (TEMPORAL PPR) PERFORMANCE ON FULL GRAPH

Class	Precision	Recall	F1-score
Licit	0.9929	0.2439	0.3916
Illicit	0.0122	0.8432	0.0241

The classification performance is reported in Table V.

The overall accuracy is 0.2504 and ROC-AUC is 0.5411.

The corresponding confusion matrix is given in Table VI.

TABLE VI
CONFUSION MATRIX (ZEBRA)

	Predicted licit	Predicted illicit
True licit	17110	53054
True illicit	122	656

Despite incorporating temporal information and learned embeddings, Zebra exhibits extremely unbalanced performance.

The model achieves very high recall for illicit nodes (0.84), indicating that it successfully identifies most illicit activity. However, this comes at the cost of extremely low precision (0.012), resulting in a massive number of false positives.

This behavior suggests that temporal diffusion combined with learned embeddings tends to over-propagate signals across the graph, failing to distinguish between true illicit structures and structurally similar but benign patterns.

In particular, the model captures activity patterns related to high transaction intensity and temporal proximity, but lacks the structural selectivity required to isolate coordinated illicit behavior.

Overall, these results demonstrate that even advanced temporal graph learning frameworks are insufficient to achieve reliable detection performance at scale.

This finding reinforces the central conclusion of this work: temporal dynamics alone do not provide enough discriminative power, and effective detection of illicit activity requires explicit modeling of higher-order structural patterns.

VIII. FULL-GRAPH TEMPORAL METHODS

We evaluate several scalable approaches on the full Bitcoin transaction graph, including diffusion-based methods, PageRank variants, and temporal graph models. The results are summarized in Table VII.

TABLE VII
PERFORMANCE OF FULL-GRAPH METHODS FOR ILLICIT ACTIVITY DETECTION

Method	Precision (%)	Recall (%)	F1-score (%)
t-PPR (seeds)	14.79	50.26	22.85
PPR (degree seeds)	14.30	53.00	22.50
PPR (random seeds)	13.30	51.30	21.10
TGN	82.81	14.11	24.12
PPR (global seeds)	8.40	3.00	4.40
PPR (personalized)	5.40	2.30	3.30
Personalized t-PPR (SSL)	5.43	6.32	5.84
Zebra (SSL)	1.22	84.32	2.41

Notably, even the strongest temporal model (TGN) fails to achieve balanced performance, highlighting the limitations of purely temporal approaches at the full-graph level.

Diffusion-based approaches such as Personalized PageRank (PPR) achieve moderate recall (around 50%) but low precision (around 13–15%), indicating that while they successfully propagate illicit signals, they produce a large number of false positives.

Global and fully personalized PageRank strategies perform poorly, with both precision and recall below 10%, suggesting that global centrality measures are not informative for this task.

Temporal Flow PageRank and related SSL-based methods do not significantly improve performance, confirming that temporal weighting alone is insufficient.

The Zebra framework achieves extremely high recall (84.32%), but at the cost of very low precision (1.22%), resulting in a large number of false positives.

In contrast, Temporal Graph Networks (TGN), a state-of-the-art model for dynamic graphs, widely considered among the most powerful temporal GNN architectures, achieve very high precision (82.81%) but extremely low recall (14.11%). This indicates that TGN is highly conservative, identifying only a small subset of illicit nodes with high confidence.

Despite being a state-of-the-art approach for temporal graph learning, TGN does not provide satisfactory performance on the full Bitcoin transaction graph. More generally, none of the evaluated methods achieves a good balance between precision and recall.

These results highlight important limitations of full-graph temporal and diffusion-based approaches:

- diffusion methods capture broad signals but lack selectivity,
- temporal models capture strong signals but miss most illicit nodes,
- high-recall methods fail due to excessive false positives.

Overall, these findings suggest that relying solely on temporal dynamics or global diffusion is insufficient for illicit

activity detection. Instead, they highlight the importance of localized structural patterns, which are better captured by ego-network-based approaches.

IX. DISCUSSION AND CONCLUSION

The results of this study highlight a fundamental contrast between structural and temporal approaches for illicit activity detection in large-scale transaction graphs.

On ego-networks, higher-order structural models such as PPGNs demonstrate strong performance, particularly in terms of recall. This indicates that illicit activity is often embedded in localized and coordinated interaction patterns, such as triangles, chains, and dense ego-communities. These structures reflect underlying social organization and cannot be captured by models limited to pairwise interactions.

In contrast, full-graph approaches based on diffusion and temporal modeling fail to achieve a satisfactory balance between precision and recall. Diffusion-based methods propagate signals broadly but lack selectivity, resulting in many false positives. Temporal methods, including state-of-the-art models such as TGN, capture only a limited subset of highly distinctive patterns, leading to high precision but very low recall.

These results suggest that temporal information alone is not sufficient to characterize illicit behavior. Instead, the key signal lies in stable structural patterns that persist across local neighborhoods.

A further limitation arises from the current graph construction. The use of BFS without strict temporal constraints leads to a pseudo-temporal representation, which may partially explain the limited impact of temporal methods. Constructing truly temporal ego-networks represents an important direction for future work.

More importantly, our experiments indicate that the main bottleneck is not model expressiveness, but scalability. Highly expressive models such as PPGNs are well suited for capturing higher-order structural patterns, but their quadratic complexity limits their application to relatively small ego-networks.

We hypothesize that larger ego-networks, containing richer and more complex motif structures, would provide a more favorable setting for expressive architectures. However, scaling PPGN-like models to larger subgraphs or to the full graph remains a significant computational challenge.

Taken together, these findings suggest that the future of illicit activity detection in transaction graphs lies in structure-aware approaches that explicitly model higher-order interactions, rather than relying solely on temporal or diffusion-based signals.

Developing scalable methods capable of capturing such structural patterns at larger scales therefore represents a key direction for future research.

These findings suggest that scaling higher-order structural models, rather than further refining temporal diffusion mechanisms, is the key challenge for future research in graph-based fraud detection.

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